

# SIMONE MANTI

INFN – Laboratori Nazionali di Frascati, Italy

ORCID 0000-0003-3770-0863 | Curriculum for INFN-CSN5 Young Researchers Grant 2026

Personal contact details, birth data, photograph and signature are intentionally omitted in accordance with call no. 28978, Art. 7(h).

## SCIENTIFIC PROFILE

Experimental and computational physicist working at the interface of laboratory X-ray spectroscopy, radiation detectors and data-driven modelling. Since 2022 at INFN-LNF, my research has combined Bragg-crystal X-ray emission and absorption spectroscopy, CZT and BEGe detector simulations and event selection, atomic-structure and cascade calculations, and first-principles high-throughput materials modelling. I am co-leader of the Software Advancement Working Group of COST Action ENRICH and an instructor in X-ray-optics ray tracing with SHADOW4/OASYS2.

## WORK & RESEARCH EXPERIENCE

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| May 2022–present    | <b>Postdoctoral Researcher (Assegnista di ricerca)</b> , <i>INFN, Laboratori Nazionali di Frascati</i><br>Laboratory X-ray spectroscopy, radiation-detector studies, precision atomic physics, simulation and machine-learning methods. Member of the VOXES, SIDDHARTA-2 and VIP research activities. |
| Sep. 2021–Apr. 2022 | <b>Research Assistant</b> , <i>Technical University of Denmark (DTU)</i><br>First-principles and high-throughput modelling of two-dimensional materials and quantum point defects. Supervisor: Prof. Kristian S. Thygesen.                                                                            |

## EDUCATION

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| Awarded Sep. 2021 | <b>Ph.D. in Physics</b> , <i>Technical University of Denmark (DTU)</i><br>May 2018–Sep. 2021. Thesis: <i>Effects of lattice imperfections on the optical and electronic properties of two-dimensional materials</i> . Supervisor: Prof. Kristian S. Thygesen.                   |
| Awarded Dec. 2017 | <b>M.Sc. in Physics, 110/110 cum laude</b> , <i>University of Rome Tor Vergata</i><br>Apr. 2016–Dec. 2017. Thesis: <i>Real-time photoionization of atoms: successes and limitations of the non-equilibrium Green's-function method</i> . Supervisor: Prof. Gianluca Stefanucci. |
| Awarded Oct. 2015 | <b>B.Sc. in Materials Science, 104/110</b> , <i>University of Rome Tor Vergata</i><br>Oct. 2012–Oct. 2015. Thesis on the formation and characterization of defects on differently oriented germanium surfaces.                                                                  |

## SCIENTIFIC LEADERSHIP AND RECOGNITION

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| 2025–present | <b>Co-Leader, WG3 Software Advancement, COST Action CA24131 ENRICH</b><br>Coordinating activities on software development and standards for the European radiation-detection research network; WG leader: Dr Manuel Sanchez del Rio. |
| 2026         | <b>Instructor, ESRF–ENRICH OASYS School EOS26</b><br>Hands-on training in X-ray-optics design and ray tracing with SHADOW4/OASYS2, ESRF, Grenoble, 20–22 May 2026.                                                                   |
| 2025         | <b>PLGrid computational grant: 500,000 core-hours on the Helios supercomputer</b><br>Grant no. PLG/2025/018524.                                                                                                                      |

## RESEARCH CONTRIBUTIONS RELEVANT TO PRISM

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**Laboratory X-ray spectroscopy and VOXES.** I operate and optimize the INFN-LNF VOXES von Hamos spectrometer, including crystal/slit alignment, energy calibration, acquisition and background suppression. As first and corresponding author, I reported a one-hour multi-element Cu–Ni–Zn XES measurement with sub-10-eV FWHM resolution near 8 keV and an average 30% reduction in peak-centre uncertainty after background selection. A 2026 follow-up documents the ED-XRF monitor, liquid-sample handling and mechanical compatibility with laboratory transmission XAS.

**MITIQO and quantitative XRF.** I have responsibility for the implementation of MITIQO, a Regione Lazio research project on trace metals and oxidation states in edible liquids. My contributions include WD- and ED-XRF measurements, experimental optimization, signal-to-background improvement, construction of elemental-composition datasets, principal-component analysis and machine-learning classification of wines by geographical origin.

**Semiconductor detectors, Monte Carlo and data analysis.** For CZT detectors at DAΦNE, I use detailed Geant4 simulations to generate labelled signal/background datasets and develop supervised event classifiers; this work was presented at XGRADE 2026. For the VIP experiment, I co-developed and validated machine-learning event selection for a BEGe detector (co-first and corresponding author, JINST 2026). These activities cover detector response, spectral and waveform features, pile-up/noise rejection and quantitative performance assessment.

**Precision atomic and quantum X-ray spectroscopy.** Within SIDDHARTA-2, I perform multi-configuration Dirac–Fock calculations and develop atomic-cascade simulations for kaonic atoms, connecting calculated transition energies and yields to measurements. I am first and corresponding author of the 2026 Physical Review A test of bound-state QED with kaonic neon. Within VIP, I contribute to first-principles transition calculations and X-ray tests of the Pauli Exclusion Principle and dynamical-collapse models; I am a corresponding author of the related 2024 Physical Review Letters article.

**First-principles materials modelling and high-throughput workflows.** My doctoral research used density-functional theory, automated workflows and databases to predict structural, electronic and optical properties of 2D materials. Contributions include dynamical-stability classification, the C2DB ecosystem and the QPOD database of more than 1,900 defect systems. This work produced first/corresponding-author and equal-contribution publications in *npj Computational Materials*, together with an article in *ACS Nano*.

## TECHNICAL EXPERTISE

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| <b>X-ray methods</b>          | Laboratory XRF, high-resolution XES and developing XAS workflows; von Hamos geometry; Bragg crystals; X-ray tubes; motorized slits and stages; energy calibration; background subtraction and spectral fitting.                |
| <b>Detectors and analysis</b> | CZT, BEGe, silicon PIN and position-sensitive detectors; detector-response modelling; spectral/waveform analysis; event classification; rate, efficiency, resolution and background studies.                                   |
| <b>Simulation</b>             | SHADOW4/OASYS2 ray tracing; Geant4 radiation transport; multi-configuration Dirac–Fock atomic calculations; atomic-cascade Monte Carlo; density-functional theory and high-throughput materials workflows.                     |
| <b>Scientific computing</b>   | Python scientific computing; supervised and unsupervised machine learning; principal-component analysis; reproducible data-processing and simulation workflows; technical instruction and collaborative software coordination. |

## SELECTED RESEARCH VISITS AND SCHOOLS

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- Wigner Research Centre for Physics, Budapest, Jan. 2024 (scientific visit with Prof. Lajos Diósi).
- Sorbonne Université, Paris, Dec. 2023 (scientific visit with Prof. Paul Indelicato).
- FTMC, Vilnius, Sep.–Dec. 2019 (Ph.D. external research stay with Dr Audrius Alkauskas).

## PUBLICATIONS MOST RELEVANT TO THE PROPOSAL

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The five peer-reviewed articles below document the core competences supporting the proposal. The applicant's name is shown in bold; \* denotes corresponding authorship and † equal contribution.

1. **S. Manti**<sup>\*</sup> and A. Scordo, "Recent Developments of the VOXES Von Hamos X-Ray Spectrometer for Laboratory XES and XAS Studies," *Engineering Innovations* **19**, 23–29 (2026). doi:10.4028/p-rQpe9Z
2. **S. Manti**<sup>\*</sup>, F. Napolitano, A. Clozza, C. Curceanu, G. Moskal, K. Piscicchia, D. Sirghi and A. Scordo, "Enhancing Performances of the VOXES Bragg Spectrometer for XES Investigations," *Condensed Matter* **9**, 19 (2024). doi:10.3390/condmat9010019
3. **S. Manti**<sup>†</sup>, S.H. Yip<sup>†</sup> et al., "Machine-learning-enhanced event selection for BEGe detectors in the VIP experiment," *Journal of Instrumentation* **21**, P01036 (2026). doi:10.1088/1748-0221/21/01/P01036
4. **S. Manti**<sup>\*</sup>, F. Sgaramella, L. Abbene et al. (SIDDHARTA-2 Collaboration), "Precision test of bound-state QED at intermediate  $Z$  with kaonic neon," *Physical Review A* **113**, 022815 (2026). doi:10.1103/rry5-tdqb
5. K. Piscicchia, S. Donadi, **S. Manti**<sup>\*</sup>, A. Bassi, M. Derakhshani, L. Diósi and C. Curceanu, "X-Ray Emission from Atomic Systems Can Distinguish between Prevailing Dynamical Wave-Function Collapse Models," *Physical Review Letters* **132**, 250203 (2024). doi:10.1103/PhysRevLett.132.250203

## ADDITIONAL SELECTED PUBLICATIONS AND PREPRINTS

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6. F. Artibani, **S. Manti** et al., "Kaonic Copper and Fluorine Absolute Yields Measurement with a CZT-based Detection System at DAΦNE," preprint (2026). arXiv:2605.10477
7. **S. Manti**<sup>\*</sup> et al., "Testing the Pauli Exclusion Principle across the Periodic Table with the VIP-3 Experiment," *Entropy* **26**, 752 (2024). doi:10.3390/e26090752
8. **S. Manti**<sup>\*</sup>, M.K. Svendsen, N.R. Knøsgaard, P.M. Lyngby and K.S. Thygesen, "Exploring and machine learning structural instabilities in 2D materials," *npj Computational Materials* **9**, 33 (2023). doi:10.1038/s41524-023-00977-x
9. F. Bertoldo<sup>†</sup>, S. Ali<sup>†</sup>, **S. Manti**<sup>†</sup> and K.S. Thygesen, "Quantum point defects in 2D materials – the QPOD database," *npj Computational Materials* **8**, 56 (2022). doi:10.1038/s41524-022-00730-w
10. S. Ali, F.A. Nilsson, **S. Manti**, F. Bertoldo, J.J. Mortensen and K.S. Thygesen, "High-Throughput Search for Triplet Point Defects with Narrow Emission Lines in 2D Materials," *ACS Nano* **17**, 21105–21115 (2023). doi:10.1021/acsnano.3c04774
11. **S. Manti**<sup>\*</sup> et al., "Detecting Iron Oxidation States in Liquids with the VOXES Bragg Spectrometer," preprint (2023). arXiv:2310.06062

## TEACHING, SUPERVISION AND OUTREACH

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- Teaching Assistant, *Electronic Structure Methods in Materials Physics*, DTU, Spring 2019; and *Condensed Matter Physics and Nanoscale Materials Physics*, DTU, Fall 2018.
- Supervised students: Carlo Zamponi, Alex Tubby, Siu-Hung Yip, Michaela Lizardou and Luiz Gustavo Fernandes.
- M.Sc. thesis co-supervisor: Tobia Vecchiatto Montorio, University of Padua, *New Force Mediators and Their Impact on Exotic Atoms*, academic year 2025/2026.
- Contributor to INFN-LNF educational and outreach activities for the B. Touschek high-school programme and the "Incontri di Fisica" Winter Stages for science teachers.

## PEER REVIEW

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Reviewer for *Journal of Instrumentation (JINST)*, *Scientific Reports* and *Few-Body Systems*.

## LANGUAGES

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|---------|----------------------------------|
| Italian | Native proficiency               |
| English | Professional working proficiency |
| Danish  | Elementary proficiency           |

## SELECTED INVITED TALKS, LECTURES AND CONTRIBUTIONS

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| 22–26 Jun. 2026     | <b>Lecturer, 2nd BridgeQG Training School</b><br>“Testing quantum gravity with underground experiments.”                                                                                                                    |
| 14–19 Jun. 2026     | <b>Talk, European Conference on X-Ray Spectrometry (EXRS 2026), Catania</b><br>“VOXES: a HAPG-Based Von Hamos Spectrometer for High-Resolution X-Ray Spectrometry of Extended Sources.”                                     |
| 26 Mar. 2026        | <b>Oral contribution, XGRADE 2026, University of Palermo</b><br>“Machine Learning background suppression for CZT detectors using Monte Carlo simulations.”                                                                  |
| 3–5 Dec. 2025       | <b>Invited talk, Exotic Atoms: Fundamental Aspects, Applications and Advances in Radiation Detectors, Zagreb</b><br>“Testing Bound-State QED in Strong Fields with Kaonic Atoms.”                                           |
| 28 Oct.–1 Nov. 2025 | <b>Talk, 16th European Research Conference on Electromagnetic Interactions with Nucleons and Nuclei (EINN 2025), Paphos, Cyprus</b><br>“AI-Enhanced BEGe Detectors for Low-Energy X-ray Collapse Model Tests.”              |
| 8–10 Oct. 2025      | <b>Talk, Perspectives at the Bellotti IBF Workshop, LNGS, Assergi</b><br>“Experiments at Bellotti IBF to Reach beyond the Standard Model.”                                                                                  |
| 24–26 Sep. 2025     | <b>Invited talk, 2nd Symposium on New Trends in Nuclear and Medical Physics, Krakow</b><br>“High-Precision X-ray Spectroscopy: from Kaonic Atoms to Societal Applications.”                                                 |
| 24–29 Aug. 2025     | <b>Invited talk, Workshop on Standard Model and Beyond, Corfu</b><br>“High-Precision Spectroscopy of Kaonic Atoms at SIDDHARTA-2: Probing Fundamental Interactions.”                                                        |
| 16–20 Jun. 2025     | <b>Talk and Conference Chair, High Precision X-Ray Measurements 2025, INFN-LNF</b><br>“XRF Spectroscopy with VOXES: Techniques, Optimization and Applications.”                                                             |
| 30 Sep.–4 Oct. 2024 | <b>Talk, KAMPAI Workshop, ECT*, Trento</b><br>“Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations and Monte Carlo Simulations.”                                 |
| 1–4 Jul. 2024       | <b>Invited talk, Advances in the Investigation of Weak and Strong Interactions, Bucharest, Romania</b><br>“Kaonic Atom Properties and Cascade Models: Exploring Strong Interactions through First-Principles Calculations.” |
| Jun. 2024           | <b>Invited talk, A Modern Odyssey: Quantum Gravity Meets Quantum Collapse, ECT*, Trento</b><br>“Cancellation Effects in the Spontaneous Collapse Rate at Low-Energy Limit.”                                                 |
| Nov. 2023           | <b>Invited talk, ALPACA Workshop, ECT*, Trento</b><br>“Machine Learning in SIDDHARTA-2: the Monitoring Challenge.”                                                                                                          |
| 19–23 Jun. 2023     | <b>Talk, High Precision X-Ray Measurements 2023, INFN-LNF</b><br>“Investigating Metals’ Content and Oxidation States in Edible Liquids using XRF Analysis with the VOXES X-Ray Spectrometer.”                               |
| Oct. 2022           | <b>Invited talk, EXOTICO Workshop, ECT*, Trento</b><br>“Cascade Models for Atomic Transitions in Kaonic Atoms.”                                                                                                             |